



Homework Assignment #3

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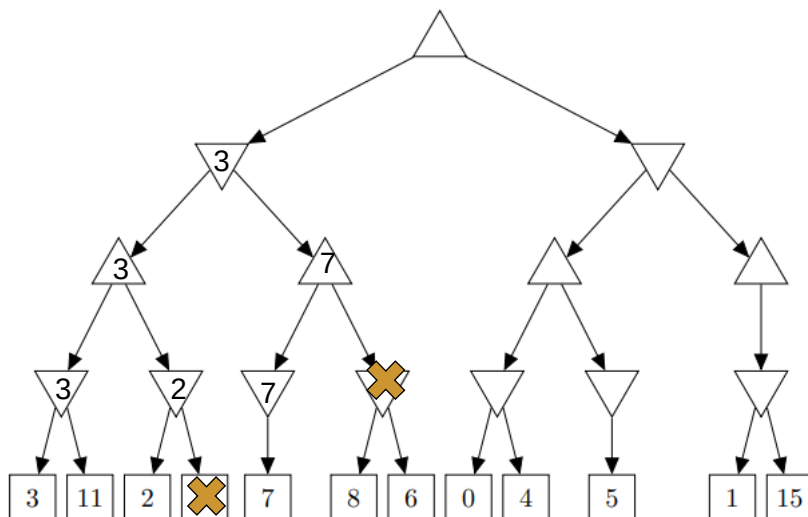
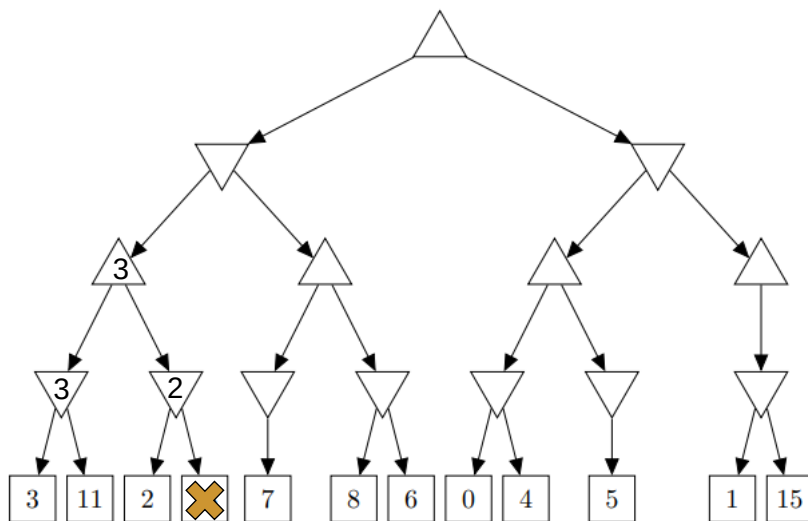
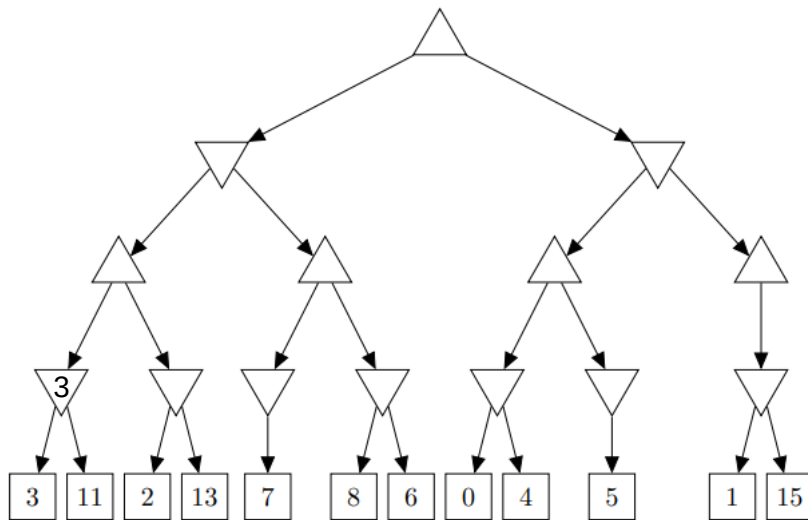
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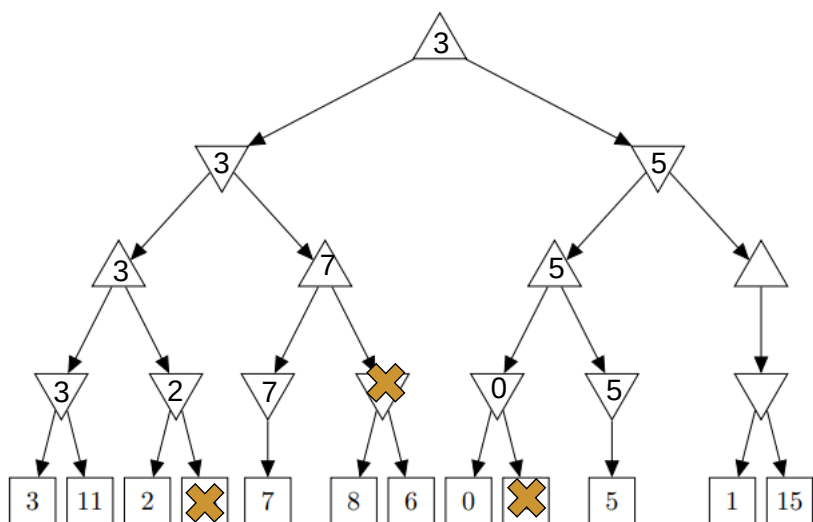
CMP 333: Artificial Intelligence

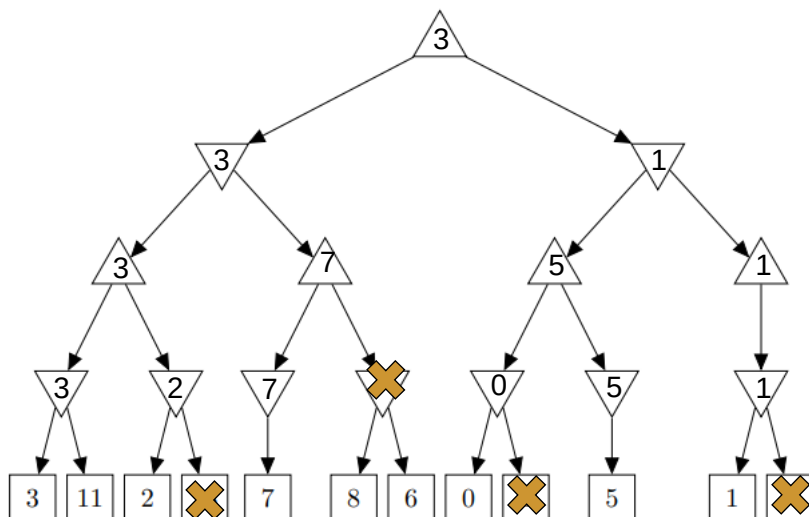
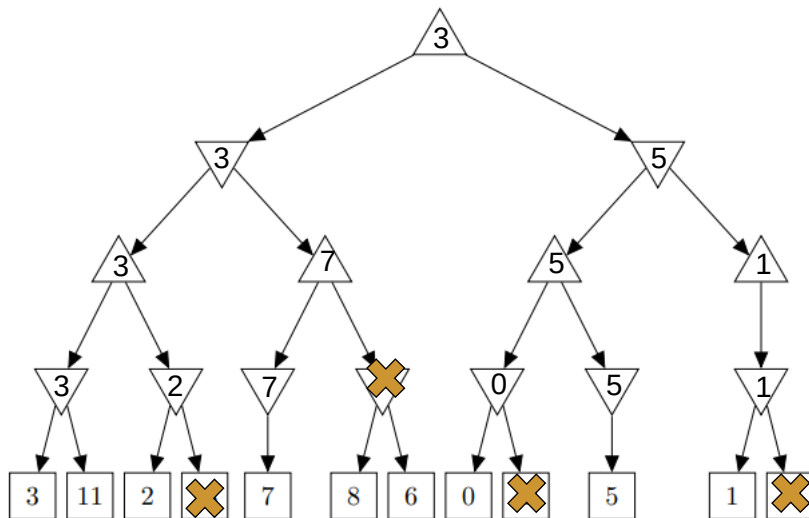
Omar Arif

July 14th, 2024

1. In the game tree below, each leaf node contains a utility value for a game state. Your task is to perform a minimax search with alpha-beta pruning to determine the optimal strategy for the maximizing player (the first player). Remember that a triangle pointing up represents the MAX value.
 - a. Show all steps for how the results were obtained. [4]







Explored List: 3, 11, 3, 2, 2, 3, 7, 7, 7, 3, 0, 0, 5, 5, 5, 1, 1, 1, 1, 3

Pruned Action: $2 \leq 3$, $7 \geq 3$, $0 \leq 3$, $1 \leq 3$

- b. Mark the nodes that were pruned [4]

Pruned Nodes: 13, 8 & 6, 4, 15

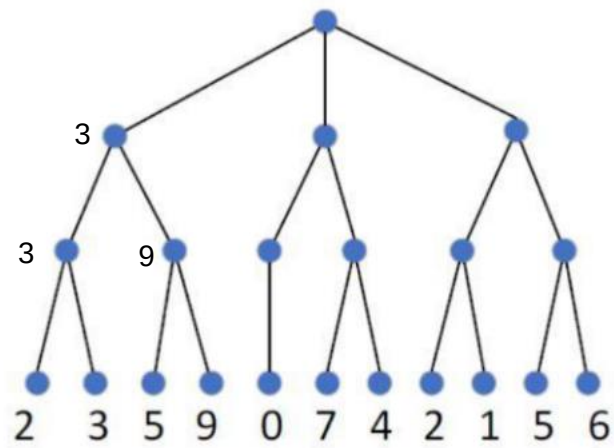
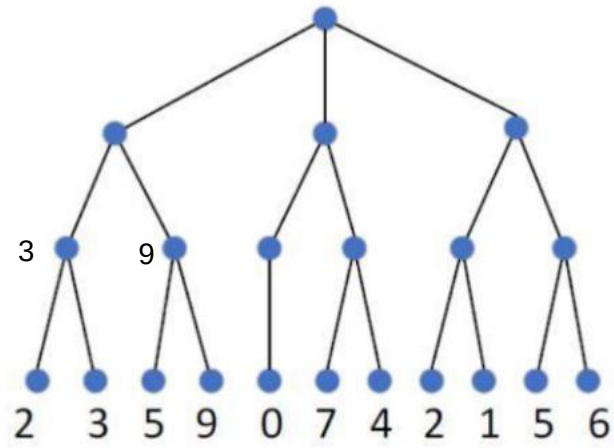
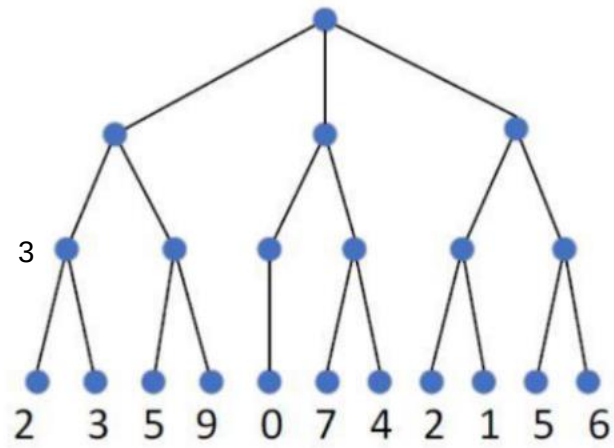
- c. What's the maximum value at the root node? [2]

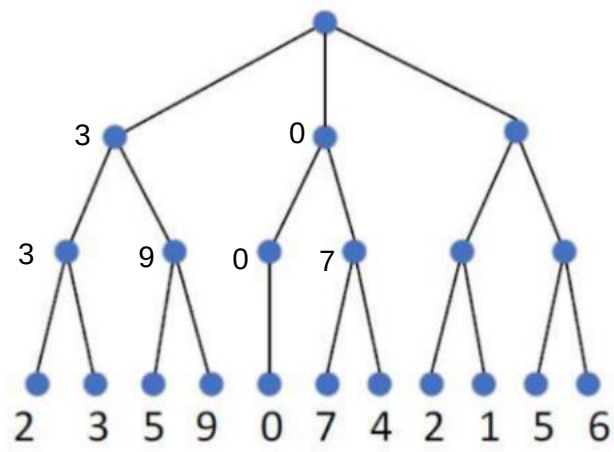
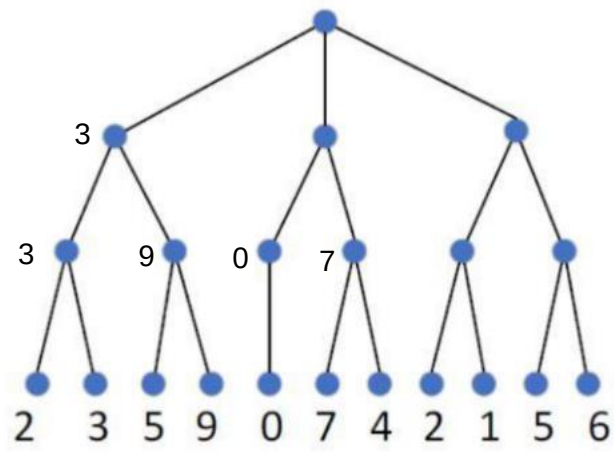
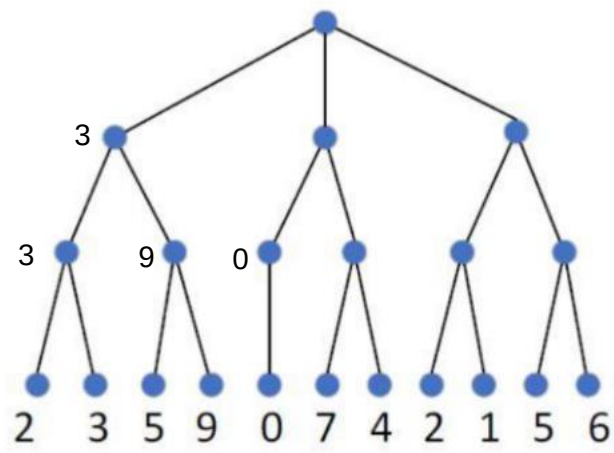
The maximum value is 3.

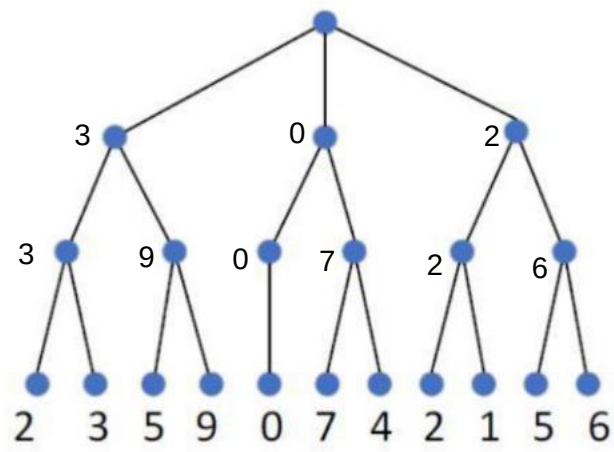
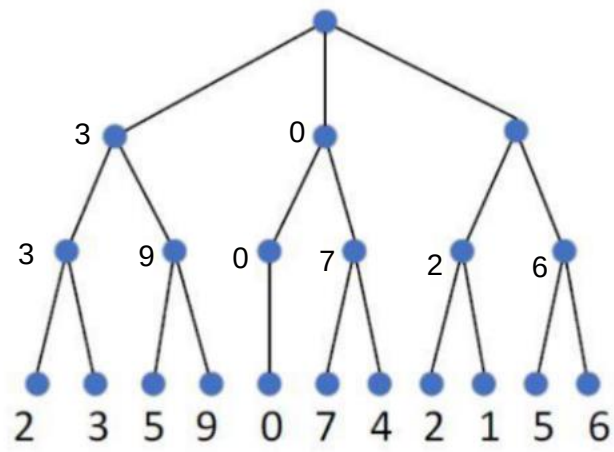
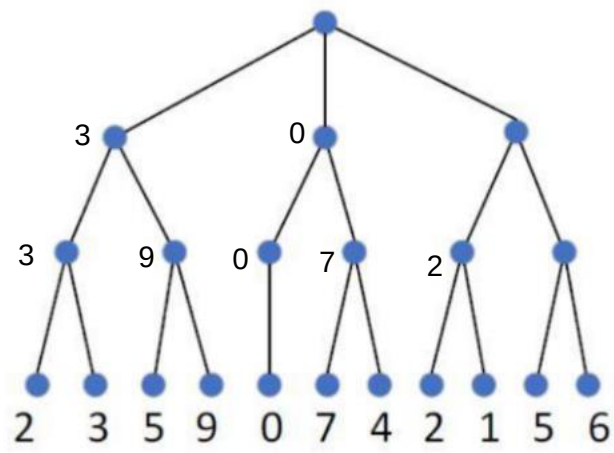
2. Use the Minimax algorithm to

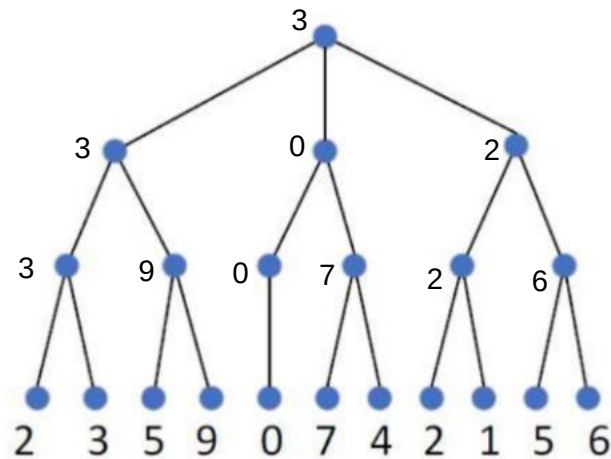
- a. Show all steps for how the results were obtained. [4]

b. Compute the value at each node for the game tree below [4]



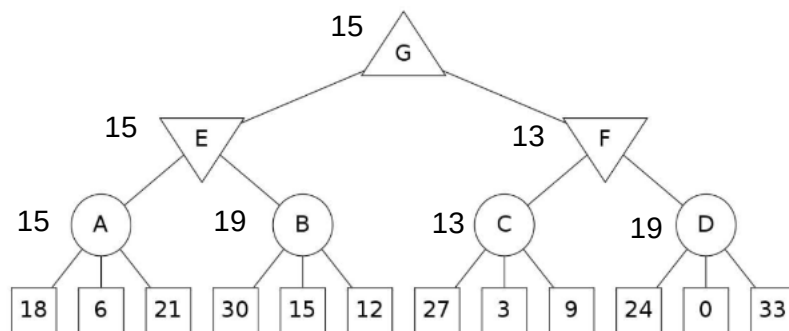






- c. What's the MAX value at the root node? [2]
The maximum value is 3.

3. Consider the game tree shown below. Similar to the previous problem, triangles pointing up represent choices for the maximizing player; triangles pointing down represent choices for the minimizing player. Circular nodes represent chance nodes, where each possible action has an equal probability. Square nodes at the bottom are leaf nodes. Assuming both players act optimally, carry out the expectiminimax search algorithm. Fill in the values for the lettered nodes in the boxes below the tree. [7]



Chance Nodes: -

A: $(18 + 6 + 21) / 3 = 15$

B: $(30 + 15 + 12) / 3 = 19$

C: $(27 + 3 + 9) / 3 = 13$

D: $(24 + 0 + 33) / 3 = 19$

1. Consider the following game tree where one of the leaf nodes has an unknown payoff x. Player 1 moves first and attempts to maximize the value of the game.

- a. Assume Player 2 is a minimizing agent (and Player 1 knows this). For what values of x is Player 1 guaranteed to choose Action 1? [2]

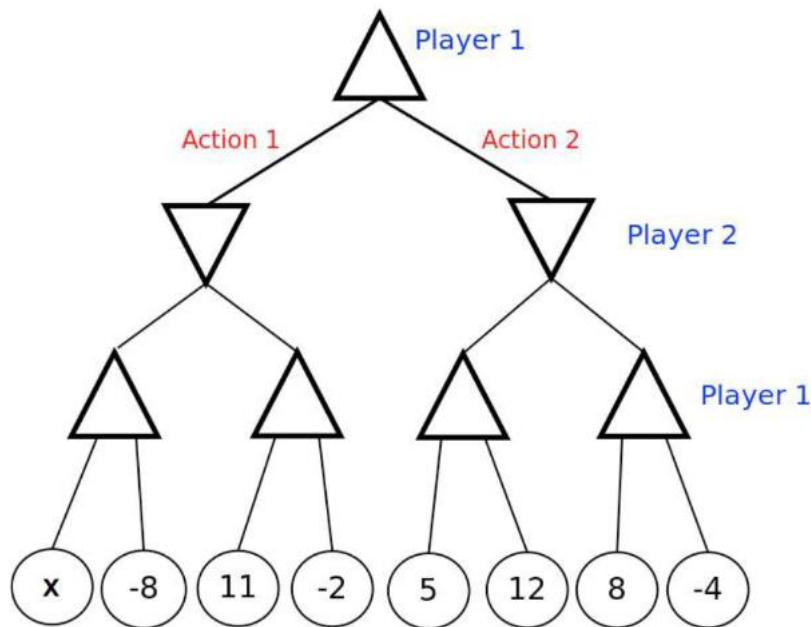
Values of x could either be 9 or 10 ($8 < x < 11$).

- b. Assume Player 2 chooses actions at random with each action having equal probability (and Player 1 knows this). For what values of x is Player 1 guaranteed to choose Action 1? [2]

Values of x that are greater than 9 ($x > 9$).

- c. Is it possible to have a game where the minimax value is strictly larger than the expectimax value? [2]

No, it is not possible. Since, the minimax algorithm can only have values 9 or 10 at the root node. However, the expectimax algorithm can have values that are > 9 (till $+\infty$). There is no possibility of having a game where the minimax value is larger than the expectimax value.



Short Answer Questions:

- List the key characteristics that differentiate games from standard search problems. [4]
 - It is an adversarial environment (playing a game against an opponent).
 - It is non-deterministic (prone to randomness of the player's decision or game).
 - It involves turn taking by each player.
 - Evaluated using utility functions or payoffs.
- You are running the minimax algorithm on a large game tree. However, you know that the minimax value is between $x - y$ and $x + y$, where x is a real number and y is a small real number. You know nothing about the minimax values at the other nodes. Describe briefly but

precisely how to modify the alpha-beta pruning algorithm to take advantage of this information. [2]

Since we know the range of the minimax value instead of setting alpha and beta to ∞ and $-\infty$. Initialize alpha to the lower bound $x - y$ and beta to the upper bound $x + y$. This immediately narrows the search space and potentially allows for more pruning for an efficient search process.

True or False Questions. If false, explain why:

1. With alpha-beta pruning, a poor choice in the order of expanding child nodes can lead to an incorrect value at the root node. [3]

False - Alpha-beta pruning is an optimized technique for the minimax algorithm. The final value of the root node is not affected by the order in which child nodes are expanded. However, the order matters for the efficiency of the alpha-beta pruning algorithm. A good choice in the order of expanding child nodes can lead to more pruning and a faster search. While a poor choice can lead to less pruning and a slower search.

2. When using expectimax to determine a policy, multiplying all the leaf node values by 10 can result in a different optimal policy. [3]

False - In expectimax, the optimal policy is determined by the relative values of the leaf nodes. Multiplying all the leaf node values by a constant factor of 10 does not change their relative values. Therefore, the optimal policy remains the same because the expected values at decision nodes will be scaled by the same factor, preserving the order of preferences among different actions.